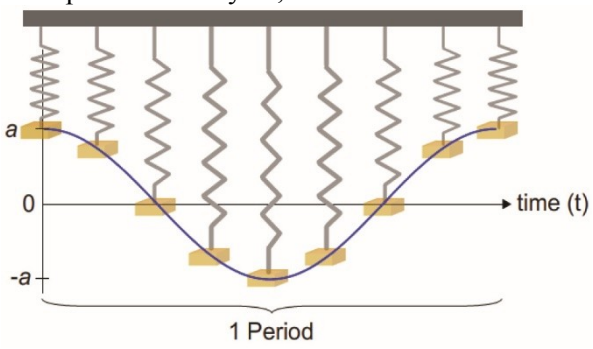
Model Simple Harmonic Motion

Consider a block moving up and down attached to a spring. In the absence of other forces, called damping forces, the block will move up and down indefinitely in a motion called *simple harmonic motion*.

In the figure, a block at the end of the spring is displaced a units upwards from equilibrium position and then released.

In the absence of damping, the model $d = a\cos\omega t$ describes the vertical displacement d as a function of the time t after release.

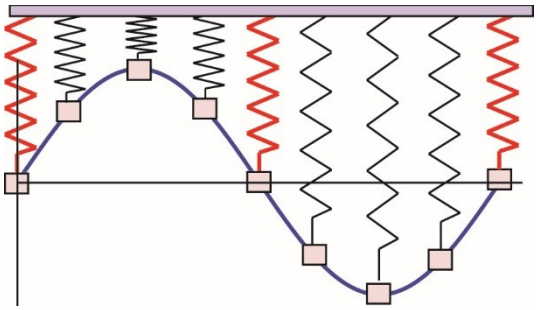
The period P is the time required to complete one up-and-down cycle, $P = \dfrac{2\pi}{\omega}$.



If the displacement d of an object at time t is given by $d = a\sin\omega t$ or $d = a\cos\omega t$ then the object is said to have **simple harmonic motion**.

Amplitude is $|a|$ (maximum displacement) Period is $P = \dfrac{2\pi}{\omega}$ (time for one complete cycle)

Frequency is $f = \dfrac{\omega}{2\pi}$ (number of cycles per unit time)



The difference between the models $d = a\cos\omega t$ (above) and $d = a\sin\omega t$ (at the left) is the position of the object at $t = 0$. Initially, if $d = 0$,

$$d = a\cos(\omega\cdot 0) = a \qquad d = a\sin(\omega\cdot 0) = 0$$

Use sine if the object "starts" at equilibrium.
Use cosine if the object "starts" at maximum displacement.

If d is something other than 0, a, or $-a$ initially, then either function can be used with a phase shift.

Note the relationship between frequency and period: $P = \dfrac{2\pi}{\omega}$ $f = \dfrac{\omega}{2\pi} = \dfrac{1}{P}$

If the period is 5 seconds, then the frequency is $\dfrac{1}{5}$ cycle per second.

If the period is 0.25 seconds, then the frequency is___4____ cycles per second. $f = \dfrac{1}{0.25}$

The unit of measurement assigned to frequency is "cycles per second" or Hertz (abbreviated Hz).

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The relationship $\omega = \dfrac{2\pi}{P}$ implies that $\omega = 2\pi f.$ Therefore, simple harmonic motion can also be modeled by $d = a\cos 2\pi ft$ and $d = a\sin 2\pi ft.$

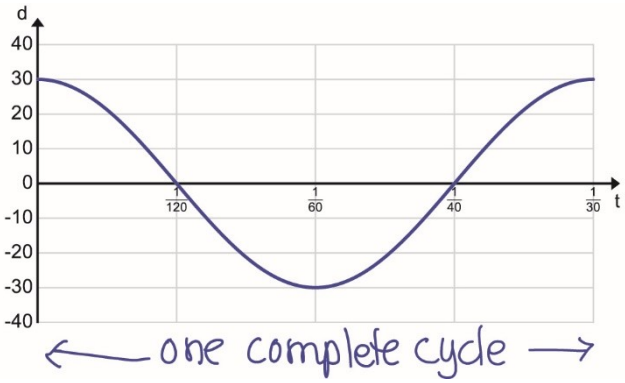
Suppose that an object moves in simple harmonic motion with displacement d (in inches) at time t (in seconds). Determine the amplitude, period, and frequency.

1. $d = 30\cos 60\pi t$

$f = \dfrac{\omega}{2\pi} = \dfrac{1}{P} = 30$ (cycles in 1 sec.)

$A =$ 30 (inches)

$P = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{60\pi} = \dfrac{1}{30}$ (second)



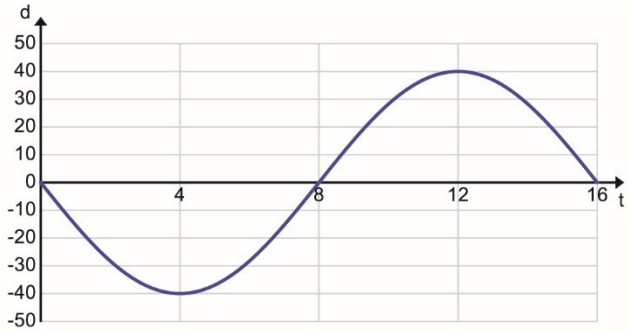
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2. $d = -40\sin\dfrac{\pi}{8}t$

$f = \dfrac{\omega}{2\pi} = \dfrac{1}{P} = \dfrac{1}{16}$ (cycle per sec or Hz)

$A =$ 40 (inches)

$P = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{\pi/8} = 16$ (seconds)



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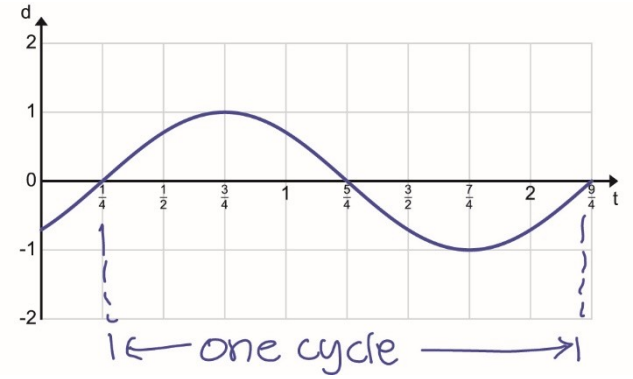
3. $d = \sin\left(\pi t - \dfrac{\pi}{4}\right)$

$P = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{\pi} = 2$ (sec)

$A =$ 1 inch

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$$f = \frac{\omega}{2\pi} = \frac{1}{P} = \frac{1}{2}(\text{Hz})$$



$$\text{phase shift} = \frac{\frac{\pi}{4}}{\pi} = \frac{1}{4} \text{ to the right}$$

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4. A block hangs on a spring attached to the ceiling and is pulled down 8 inches below its equilibrium position. After release, the block makes one complete up-and-down cycle in 5 sec and follows simple harmonic motion.

a. What is the period of motion? b. What is the frequency? c. What is the amplitude?

one cycle in 5 sec

$P = 5$ sec

$f = \frac{1}{P} = \frac{1}{5} = 0.2$

0.2 cycles/sec

8 in. below

$A = 8$ inches

d. Write a function to model the displacement d (in inches) as a function of the time t (in seconds) after release. Assume that a displacement above the equilibrium position is positive.

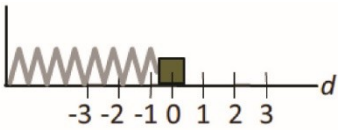
$P = \frac{2\pi}{\omega}$

$5 = \frac{2\pi}{\omega} \rightarrow \omega = \frac{2\pi}{5}$

$d = -8\cos\frac{2\pi}{5}t$

starts below equilib.

Suppose that an object is attached to a horizontal spring subject to the given conditions. Find a model for the displacement d as a function of the time t.



cosine

5. At time $t = 0$ the initial displacement is 4 inches, the amplitude is 4 inches, and the period is 2.8 seconds.

pos

$2.8 = \frac{2\pi}{\omega}$

$\omega = \frac{2\pi}{2.8} = \frac{5\pi}{7}$

$d = 4\cos\frac{5\pi}{7}t$

sine

6. At time $t = 0$, the initial displacement is 0 inches (moving to the right), the amplitude is 16 centimeters, and the frequency is 4 Hz.

positive

$f = \frac{\omega}{2\pi}$

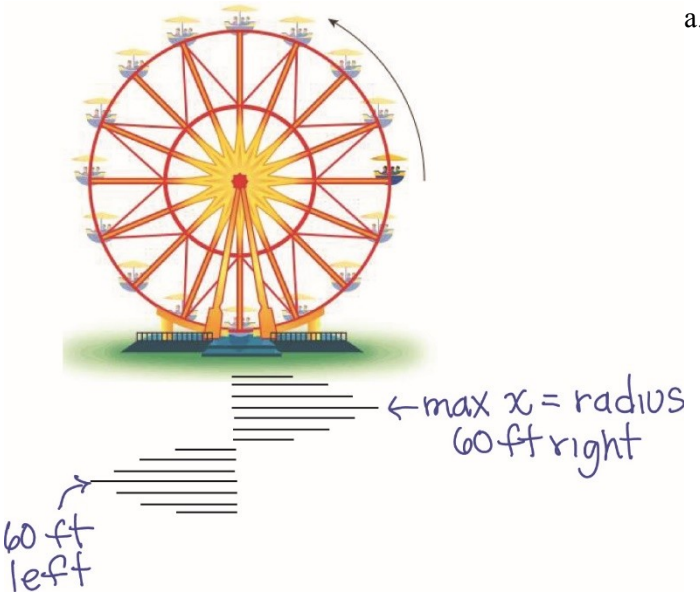
$\frac{\omega}{2\pi} = 4$

$\omega = 8\pi$

$d = 16\sin 8\pi t$

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7. A Ferris wheel is 120 ft in diameter with its lowest point 2 feet off the ground. The wheel makes one full rotation counterclockwise in 2 minutes. Suppose that someone is seated at the lowest point on the wheel and is the last passenger to be loaded when the wheel starts.



a. Write a model representing the last passenger's horizontal position x (in feet) relative to the center of the Ferris wheel, t minutes after the ride starts.

max deviation from $x = 0$ is $x = 60$

Period = 2 minutes $\quad \dfrac{2\pi}{\omega} = 2$

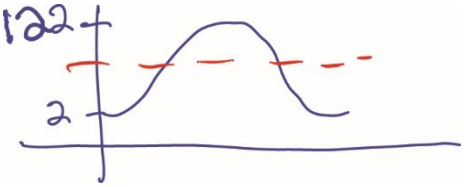
$$\omega = \frac{2\pi}{2} = \pi$$

$$x = 60 \sin \pi t$$

b. Write a model representing the last passenger's height y (in feet) above ground level, t minutes after the ride starts.

height begins 2ft off ground
highest 122 ft

$$\text{vertical shift} = \frac{2+122}{2} = 62 \text{ ft.}$$



$$P = \frac{2\pi}{\omega}$$

$$\frac{2\pi}{\omega} = 2$$

$$\omega = \pi$$

$$\frac{122-2}{2} = 60, \text{ Amplitude} = 60$$

starts at lowest point

$$y = -60 \cos \pi t + 62$$

Interpret Damped Harmonic Motion

Damped Harmonic Motion The displacement d of an object oscillating under damped harmonic motion at time t is given by

$$d = ae^{-ct} \cos \omega t \text{ or } d = ae^{-ct} \sin \omega t \text{ for } c > 0.$$

a is the displacement of the object at $t = 0$.

c is a constant called the *damping factor* and is related to the damping force opposing motion. (If $c = 0$, the model is for *simple* harmonic motion.)

ω is related to the period P and frequency f by $P = 2\pi/\omega$ and $\omega = 2\pi f$.
(*Period* in this context is the time required for one complete back-and-forth cycle.)

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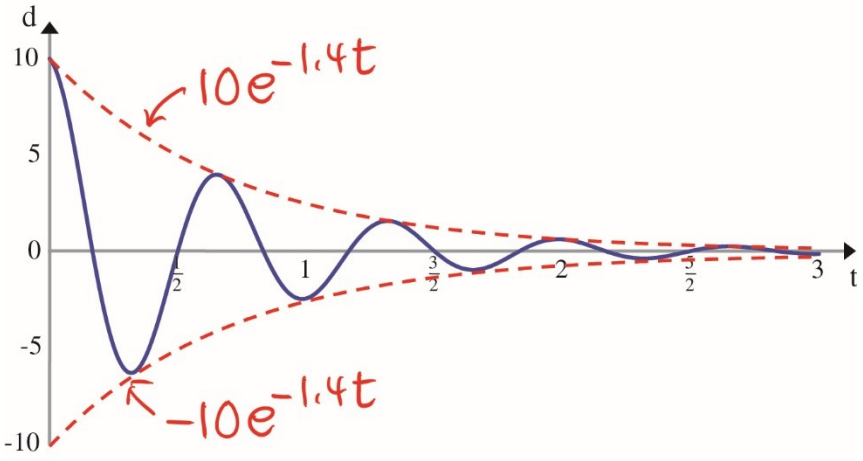
8. The displacement (in inches) of an object oscillating under damped harmonic motion at time t (in seconds) is given by $d = 10e^{-1.4t}\cos 3\pi t$.

Identify the displacement of the object at time $t = 0$, the damping factor, and the period.

$d(0) = 10e^{-1.4(0)}\cos 3\pi(0) = 10$ inches — initial displacement

damping factor: $c = 1.4$

$$\text{period} = \frac{2\pi}{\omega} = \frac{2\pi}{3\pi} = \frac{2}{3}$$



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